

Aakriti Shah

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Education

- MS University of Southern California**, Computer Science Aug 2024 - **Exp.** May 2026
- GPA: 3.75 – *Magna Cum Laude*
 - Coursework:** Trustworthy Machine Learning, Natural Language Processing, Deep Learning and Its Applications, Algorithms, Innovation for Defense Applications, Text as Data (NLP and LLMs)
- BA Rollins College**, Computer Science & Business Management (*Honors*) Sept 2020 – Dec 2023
- GPA: 3.72 – *Magna Cum Laude*
 - Coursework:** Operating Systems, Algorithms, Software Engineering, Computer Architecture, Data Science, Discrete Math, Software Engineering

Technical Skills

Programming: Python, SQL, Java, JavaScript, R, Assembly (ARM), C, C++, XML, Dart, C#, LaTeX, Swift
ML/DL Frameworks: PyTorch, TensorFlow, Hugging Face, scikit-learn, pandas, NumPy
Tools & Platforms: Git, Docker, Jupyter, Tableau, Google Cloud Platform, MongoDB, Flask, React, Node.js
Research Skills: Experiment design, data annotation, literature review, statistical analysis, scientific writing

Experience

- NSF IRES Research Fellow**, Northumbria University – Newcastle, UK Summer 2026
- Selected for the NSF-funded Citizen-Centered AI Summer Research Program, conducting human-centered AI research in an international, interdisciplinary setting, with the theme of **Identity, Privacy, Security, and Information**.
 - Collaborating with researchers from the U.S., U.K., Germany, and Portugal on sociotechnical AI challenges at the intersection of community, design, and society.
- Group Project Manager and Developer**, USC × Defense Health Agency Jan 2025 – May 2025
[Project](#) [Paper](#)
- Collaborated with Dr. David Brody of Walter Reed National Military Medical Center to design and build a three-component system, an AI triage app, drone resupply optimizer, and portable eddy current hemorrhage detector, aimed at reducing preventable TBI deaths in contested battlefield environments.
 - Acted as project manager on this team of students.
- Research Assistant**, ARKAI Lab - Dr. Thai Le (Computer Science, Indiana University) May 2025 – Ongoing
[Project](#) [Paper](#)
- Investigating **targeted unlearning** in LLMs as a **privacy-preserving** mechanism, analyzing how selectively removing sensitive information mitigates risks of **unauthorized data exposure**.
 - Examining **knowledge entanglement** in unlearned LLMs to assess **information leakage** vulnerabilities, evaluating resistance to **adversarial extraction attacks** and **membership inference**.
 - Developing evaluation frameworks to assess **unlearning robustness** against adversarial threats, informing the design of **privacy-compliant** and **attack-resilient** LLMs.
- Research Assistant**, AIMed Lab - Dr. Rakesh Shiradkar (Computer Science, Indiana University) May 2025 – Ongoing
[Project](#)
- Conducting computational pathology research on prostate **cancer detection** through automated tissue compartment and Gleason pattern segmentation in H&E-stained whole slide images.
 - Extracted patches from and annotated glands from H&E-stained whole slide images using

QuPath.

- Developed **novel postprocessing pipeline** incorporating lumen reconstruction to improve segmentation accuracy of tissue compartments (stroma, lumen, epithelium), achieving average Dice score of 0.895.
- Implementing and evaluating **deep learning and vision foundation architectures** (UNet, nnUNet, UNI) for robust multi-class segmentation across diverse histopathological datasets.
- Extending research to Gleason pattern grading (patterns 3, 4, 5) to enable deep learning automated cancer aggressiveness assessment.

Research Project, NLP and LLMs (University of Southern California)

Aug 2025 – Dec 2025

[Paper](#) 

- Used multiple unlearning algorithms (DPO, refusal tuning, and prompting) to remove political information from different LLMs (Llama-3.1-8B & Mistral-7B) to see the effects.
- These effects were measured using political compass and social psychology questionnaires (Pew Research Center Political Typology Test, Right-Wing & Left-Wing Authoritarianism, Social Dominance Orientation, and Openness to Experience in the Big Five).
- Specifically, if we remove information of politicians of a particular political party, does this shift the LLMs observed “political orientation” away from that ideology?

Research Assistant, Dr. Anita Penkova (Computer Science, University of Southern California)

Sept 2024 – Dec 2025

- Conducting **computer vision** research for early and automated detection of Diabetic Retinopathy in retinal fundus images.
- Developing and optimizing preprocessing pipelines (CLAHE, Gaussian Blur, MGA-CSG) to address limited high-quality training data and **improve model generalization** across diverse image quality conditions.

Research Project, Natural Language Processing (University of Southern California)

Jan 2025 – May 2025

[Project](#)  [Paper](#) 

- Developed Hinglish **conversational chatbot** by curating multilingual datasets with code-switched and synthetically generated training samples to address low-resource language challenges.
- **Fine-tuned instruction-tuned LLMs** for open-domain dialogue generation; evaluated outputs using syntactic and semantic metrics to assess code-switching quality and conversational coherence.

Honors Thesis, Dr. Daniel Myers (Rollins College)

Jan 2023 – Dec 2023

[Project](#)  [Paper](#) 

- **Evaluated LLM performance** on sentiment analysis using manually labeled real-world interview data from non-profit Crave, comparing human-coded emotional assessments to model-generated outputs.
- Investigated **limitations of LLMs in capturing nuanced emotional context** in low-resource, qualitative settings, providing insights for responsible deployment in social science research.

Research Intern, National Science Foundation REU – **ML in Cybersecurity**

June 2023 – Aug 2023

[Paper](#) 

- Investigated **adversarial robustness of ML models** for autonomous driving systems, analyzing vulnerabilities under perturbation attacks and evaluating transfer learning across architectures.
- Conducted interdisciplinary research, covering qualitative and quantitative methods, research ethics, and perspectives from computer science and psychology.

Technical Personal Projects

ASLearning, ML for Accessibility

Jan 2023 – May 2023

[Project](#) 

- Built mobile application using real-time object detection (TensorFlow) to facilitate communication between hearing-impaired and English-speaking users through sign language recognition.

voteSmarter, Civic Engagement Mobile App

- Developed mobile application connecting users to political candidates based on values alignment and location; integrated features for voter registration, polling locations, and candidate databases.

Feb 2024 – June 2024

[Project](#)

Dog Rescue Website

- Developed a website for a dog rescue that runs completely on donations. Their only on-line presence previously was their Facebook page, and their growing rescue needed a bigger space to spread information about their adoptable dogs.

Oct 2025 – Jan 2026

[Website](#)

Publications & Submissions

Shah, A. (presenting author), Bogale, Y., Murugan, N., Nair, V., Collins, K., Bahler, C., Feldman, M., Shiradkar, R. "Deep-learning-based segmentation of glands and stroma on digitized H&E-stained prostate whole slide images", **SPiE Medical Imaging 2026** (oral presentation).

February 2026

Accepted for Oral Presentation

Shah, A. (presenting author), Bogale, Y., Nair, V., Collins, K., Feldman, M., Bahler, C., Shiradkar, R. "Automated gleason pattern segmentation in prostate whole slide digitized pathology using foundation model feature embeddings", **Annual Meeting of the American Urological Association (AUA), 2026** (interactive poster).

May 2026

Accepted for Poster Presentation

Shah, A. (presenting author), Bogale, Y., Murugan, N., Nair, V., Collins, K., Feldman, M., Bahler, C., Shiradkar, R., "Automated segmentation of gland and stromal morphology on digitized H&E-stained prostate whole slide images", **Annual Meeting of the American Urological Association (AUA), 2026** (interactive poster).

May 2026

Accepted for Poster Presentation

Shah, A., & Le, T. (2025). "The Limits of Obliviate: Evaluating Unlearning in LLMs via Stimulus-Knowledge Entanglement-Behavior Framework" arXiv preprint [arXiv:2510.25732](https://arxiv.org/abs/2510.25732)

November 2025

In Progress

Shah, A. (2023). "Evaluating AI Sentiment Analysis." Rollins College Honors Program Theses, No. 218. Available at: scholarship.rollins.edu/honors/218/

December 2023

Honors Thesis

Teaching and Outreach

Volunteer, Amy's Angels Dog Rescue

- Volunteered to create a website for this dog rescue whose only online presence was Facebook.
- Needed a way to communicate their adoptable dogs and donation needs to their community.

Sept 2025 - Ongoing

[Website](#)

Volunteer, Viterbi Impact

- Engaging in community service projects including fire clean-up, premature baby care, and food distribution.
- Helped distribute food for people who were affected by the 2025 Southern California Wildfires and the nationwide EDT crisis.

Aug 2024 – Ongoing

Volunteer, AAPI Coming Together (ACT)

- Supporting civic outreach and cultural advocacy within the Asian American community.
- Developed **voteSmarter** to support voter education efforts and civic engagement.

May 2024 – Ongoing

Instructor, iD Tech Camps

- Taught Java, Python, C#, and game development to K-12 students, promoting computational thinking and creative problem-solving in early learners.

June 2022 – Aug 2022

Honors & Awards

Dean's List, University of Southern California (multiple semesters)

2025-2026

REU & IRES Awardee, National Science Foundation

2023 & 2026

Archibald Granville Bush Award: Excellent Standard in Academic Pursuit, Rollins College

2023